DEPARTMENT OF INDUSTRIAL & MANAGEMENT ENGINEERING

DMS201: INTRODUCTION TO OPERATIONS MANAGEMENT

END SEMESTER EAMINATION [2025-2026, SEMESTER II]

NOTE THE FOLLOWING

1) Time duration for is SIXTY (60) minutes / EIGHTY (80) for non-DAP / DAP students respectively.
2) Closed Notes/Book/Slide examination.
3) No mobiles or any other device allowed.
4) Only calculator and pen/pencil allowed
5) Exchange of calculator, pen/pencil between students not allowed.
6) Total number of questions is **TWENTY** (**20**) and the questions are
   - ❖ Exact answer (Quantitative).
   - ❖ MCQ (FIVE (05) options with ONE (01) CORRECT answer (Qualitative).
7) Marks: CORRECT is +2/Plus TWO; WRONG is −1/Minus ONE; No ANSWER is 0
8) Total mark for the end semester is **FORTY** (**20×2=40**).

From the calculation it was found that break-even point (BEP) was 9,000 units. It is given that fixed cost (FC) is Rs. 36,000/-, revenue per unit (r) is Rs. 9/-, then the variable cost per unit (v) (in Rs.) is? **ANSWER IS IN INTEGER**	
Range	4.9 to 5.1
Q_{BEP} = FC/(r – v), when P = 0, hence v = r – FC/Q_{BEP} = 9 – 36000/9000 = 9 – 4 = 5	

Demand is 260 units per 10 minutes while capacity per machine is 90 units per 15 minutes. Find the flow rate (units per hour) considering there are FIVE (05) machines each of the same efficiency and they are used simultaneously, separately and are operating parallel to each other? **ANSWER IS IN INTEGER**	
Range	1559.9 to 1560.1
Demand is 260 units per 10 minutes while Capacity per machine is 60 units per 15 minutes, we have Demand = 260×6 = 1560 units per hour Capacity = 90×4×5=1800 units per hour Flow rate = Min(Demand, Capacity) = Min(1560, 1800) = 1560 units per hour	

You are given demand rate of item (D) as 2,000 units per year; fixed cost (A) for each replenishment is Rs. 2,000/-; variable cost (v) as Rs. 4/-per unit and carrying cost (r) as 2 per year. Find the time between orders (in months) when you order EOQ number of units each time.

ANSWER IS IN INTEGER	
Range	0.59 to 0.61

$EOQ = Q^* = \sqrt{(2AD)}/\sqrt{(vr)} = \sqrt{(2\times2000\times2000)}/\sqrt{(4\times2)} = 1000$

$T^* = (Q^*/D)\times12 = (1000/2000)\times12 = 6$ months

$T^* = \sqrt{(2A)}/\sqrt{(Dvr)} = (\sqrt{(2\times2000)}/\sqrt{(2000\times4\times2)})\times12 = 6$ months

For min: $f(\boldsymbol{x})$: $-5x_1 - 4x_2 - 6x_3$

$x_1 - x_2 + x_3 \leq 20$

$3x_1 + 2x_2 + 4x_3 \leq 40$

$3x_1 + 2x_2 + 0x_3 \leq 30$

$x_1, x_2, x_3 \geq 0$

At the optimal value (minimum of $f(\boldsymbol{x}^*) = f(x_1^*, x_2^*, x_3^*)$ the corresponding value for x_2^* is?

ANSWER IS IN INTEGER

Range	14.00 to 16.00

$f(\mathbf{x}^*) = -5\times0 - 4\times15 - 6\times2.5 = -75$ and $(\mathbf{x}^*) = (x^*_1, x^*_2, x^*_3) = (0.0, 15.0, 2.5)$

MATLAB CODE

```
% 1. Define objective function coefficients

f = [-5; -4; -6];

% 2. Define linear inequality constraints (A*x <= b)

A = [+1 -1 +1;

   +3 +2 +4;

   +3 +2 +0];

b = [20; 40; 30];

% 3. Define lower bounds (x >= 0)

lb = zeros(3,1);

% 4. Call linprog solver

[x, fval] = linprog(f, A, b, [], [], lb);

% Display results
```

```
disp('Optimal solution:');

disp(x);

disp('Value of objective function:');

disp(fval);
```

For $max: f(\boldsymbol{x}): 5x_1 + 2x_2$

$x_1 + x_2 \leq 6$

$x_1 - x_2 \leq 0$

$x_1, x_2 \geq 0$

The optimal value (maximum in this case) for $f(\boldsymbol{x}^*) = f(x_1^*, x_2^*)$ is?

ANSWER IS IN INTEGER

Range	20.99 to 21.01

$(x^*_1, x^*_2) = (3, 3)$ and $f(\mathbf{x}^*) = f(x^*_1, x^*_2) = 5\times3 + 2\times3 = 21$, as

	X1	X2		RHS	Equation form
Maximize	5	2			Max 5X1 + 2X2
Constraint 1	1	1	<=	6	X1 + X2 <= 6
Constraint 2	1	-1	<=	0	X1 - X2 <= 0

Cj	Basic Variables	Quantity	5 X1	2 X2	0 slack 1	0 slack 2
Iteration 1						
0	slack 1	6	1	1	1	0
0	slack 2	0	1	-1	0	1
	zj	0	0	0	0	0
	cj-zj		5	2	0	0
Iteration 2						
0	slack 1	6	0	2	1	-1
5	X1	0	1	-1	0	1
	zj	0	5	-5	0	5
	cj-zj		0	7	0	-5
Iteration 3						
2	X2	3	0	1	0.5	-0.5
5	X1	3	1	0	0.5	0.5
	zj	21	5	2	3.5	1.5
	cj-zj		0	0	-3.5	-1.5

MATLAB CODE

%Max: 5x1 + 2x2

```
%x1 + x2 <= +6

%x1 - x2 <= 0

%x1, x2, x3 >= 0

% 1. Define Objective Function (Negative of coefficients for maximization)

f = [-5; -2];

% 2. Define linear inequality constraints (A*x <= b)

A = [+1, 1;

     1, -1];

b = [6; 0];

% 3. Bounds (x1, x2 >= 0)

lb = zeros(2,1);

% 4. Call linprog solver

[x, fval] = linprog(f, A, b, [], [], lb);

% 5.  Display Output Results

% Negate fval to get the actual maximum value

fprintf('Optimal solution:\n x1 = %.2f\n x2 = %.2f\n', x(1), x(2));

fprintf('Maximum value = %.2f\n', -fval);
```

In PERT Network a job/activity has the following completion times which are (i) Optimistic = 12 days, (ii) Most Likely = 3 weeks and (iii) Pessimistic = 6 weeks. Given this information the expected completion time for the same job/activity (in days) is?

ANSWER IS IN INTEGER

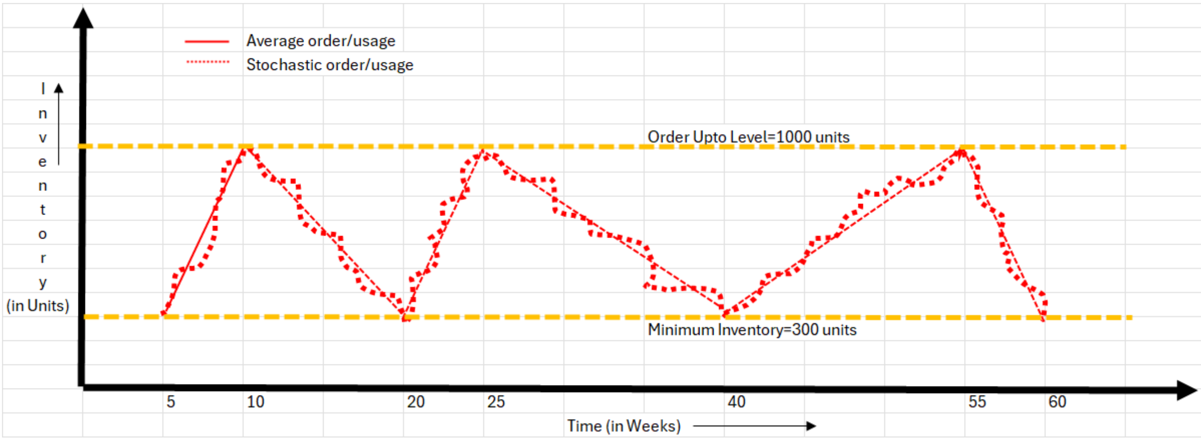
Range	22.14 to 22.20

Optimal time (t_a) = 12; Most Likely time (t_m) = 3×7=21; Pessimistic time (t_b) = 6×7=42

E(T) = Expected Time = $(t_a + 4\times t_m + t_b)/6$ = (12 + 4×21 + 42)/6 = 138/6 = 23.00 days

For the following graph (Inventory Level (**depicted along Y-axis**) versus Time (**depicted along X-axis**), what is the average inventory level (in units/week) between the starting of 5th week to beginning of 60th week?

ANSWER IS IN INTEGER



Range	649.99 to 650.01

Total inventory (05^{th} to 15^{th} week) = 300×3+ (1/2)×700×3

Total inventory (20^{th} to 35^{th} week) = 300×4 + (1/2)×700×4

Total inventory (40^{th} to 55^{th} week) = 300×4 + (1/2)×700×4

Hence average inventory (05^{th} to 55^{th} week) = [300×3 + (1/2)×700×3 + 300×4 + (1/2)×700×4 + 300×4 + (1/2)×700×4]/11 = 300 + 350 = 650 units

Mr. Gajendra Pratap has a kachori stall which is very popular in the city of Benaras. It takes about 1.0 minutes to take an order and collect the payment, while the time taken to prepare one order (i.e., 1 plate which serves two (02) kachoris and daal) is 1.5 minutes per order. Selling price per plate is Rs. 40/- while material costs per plate is Rs. 10/- and there are no fixed costs. The shop operates from 0700 hrs (IST) to 1200 hrs (IST) the day it is open. Considering the demand Mr. Pratap wishes to hire ONE (01) extra person that will reduce the per order (a plate which serves two (02) kachoris and daal) time from 1.5 minutes/90 seconds to 80 seconds. The person hired will have to be paid wage at the rate of Rs. 500/- for the total FIVE (05) days in the week the shop is open. From a purely financial perspective, what **profit** Mr. Pratap makes per week (week implies Wednesday to Monday, i.e., the shop is open SIX (06) days a week) once he hires the helper?

ANSWER IS IN INTEGER

Range	3249 to 3251

In the present process, the shop is producing at a flow rate of one order every 90 seconds which corresponds to a capacity of 1/90 plate/second, i.e., 3600×(1/90) = 40 plates/hour. Hence total number of orders in the FIVE (05) hour (0700 hrs (IST) to 1200 hrs (IST)) period is 40×5 = 200 for which the Profit = Flow Rate×(Average Price – Variable Cost) – Fixed Cost = 200×(40 – 10) = 200×30 = Rs. 6000/-

As per reduced time per order. which is 80 seconds which corresponds to a capacity of 1/80 plate/second, i.e., 3600×(1/80) = 45 plates/hour. Hence total number of orders in 5 hours is 45×5 =

225 for which the Profit = Flow Rate×(Average Price – Variable Cost) – Fixed Cost = 225×(40 – 10) = 225×30 = Rs. 6750/-. Now considering Rs. 6750 – Rs. 6000 = Rs. 750/-, the extra profit per day is 750 – 500/5. Hence per week extra profit is 5×(750 – 500/5) = Rs. (3750– 500) = Rs. 3250/-

Consider the example of the three-step airport security. The **FIRST** step, verifying ID and boarding pass, takes 30 seconds per passenger. The **SECOND** step, searching the passenger for metal objects using a metal detector, takes 10 seconds per passenger. The **THIRD** step, running the carry-on luggage through an X-ray machine, takes 60 seconds per passenger. Assume that there are many customers waiting in the process and that **EACH** of the **THREE** (**03**) employees (who are employed to man and run the three-step airport security system) gets paid Rs. 1500/- per hour. Then what is the cost of direct labor (calculate the value as Rs. per passenger)? **ANSWER IS IN INTEGER**	
Range	74 to 76
Capacity (verifying ID) = 1/30 passenger per second Capacity (metal detector) = 1/10 passenger per second Capacity (X-ray) = 1/60 passenger per second Thus X-ray is the bottle neck We know Flow rate = Min(Demand, Capacity). As many customers are waiting, hence the process is capacity-constrained, and the flow rate is given by the process capacity; which is 1/60 passenger per second or 1 passenger per minute or 60 passengers per hour. Given that we have three employees, we have to pay 3×1500 per hour = Rs. 4500/- per hour which gives us the value of cost of direct labour = 4500/60 = Rs. 75/-per passenger	

For a production process the value of fixed cost (A) is Rs. 1500/-, Demand (D) is 6000 units per year, variable cost (v) per unit is Rs. 9/- and carrying cost (r) is 2. What is the total cost (in INR) when order quantity is 20% **more** than the **economic order quantity** (**EOQ**)? **ANSWER IS IN INTEGER**	
Range	72329 to 72301
$EOQ = Q^* = \sqrt{(2AD/vr)} = \sqrt{(2\times1500\times6000)}/\sqrt{(9\times2)} = 1000$ units $Q = (1 + 20/100)\times1000 = 1.20\times1000 = 1200$ units $TRC(Q) = AD/Q + Dv + Qvr/2 = 1500\times6000/1200 + 6000\times9 + 1200\times9\times2/2 = 72300$ in INR	

A machine-paced line has **SIX** (06) stations with processing times of 30, 20, 25, 28, 22, and 24 seconds per unit, respectively. The process is **capacity-constrained** and runs on a 30 second/unit cycle time. Given this information what is the total idle time (in seconds) across all **SIX** (06) stations? **ANSWER IS IN INTEGER**

Range	30 to 32

We know the cycle time is given as 30 seconds/unit.

We now need to compute the difference between each processing time and the cycle time to get the idle time for each station.

Idle time for station 1 = Cycle time − Processing time = 30 − 30 = 0 sec per units.

Idle time for station 2 = Cycle time − Processing time = 30 − 20 = 10 sec per units.

Idle time for station 3 = Cycle time − Processing time = 30 − 25 = 5 sec per units.

Idle time for station 4 = Cycle time − Processing time = 30 − 28 = 2 sec per units.

Idle time for station 5 = Cycle time − Processing time = 30 − 22 = 8 sec per units.

Idle time for station 2 = Cycle time − Processing time = 30 − 24 = 6 sec per units.

The total idle time thus is 31 seconds, i.e., (0 + 10 + 5+ 2+ 8+ 6).

Assume Demand is 20 units per 5 minutes while Capacity per machine is 30 units per 10 minutes, and this is ONE (01) machine only. Find the flow rate/throughput (in units per hour). **ANSWER IS IN INTEGER**

Range	179 to 181

Demand is 20×12 = 240 units per 5×12 = 60 minutes (1 hour)

Capacity (ability to process) per machine is 30×6 = 180 units per 10×6 = 60 minutes (1 hour).

Hence Flow Rate or Throughput = min(20×12, 30×6) = min(240, 180) = 180 units per hour, and it is a capacity Constrained process

SBI offers customized loans. Customers have specific loan request and when they visit the bank are able to obtain a response on an average within 30 minutes. Queries for SBI loan includes five activities that must be conducted in the sequence described below.

- **Activity 1**: Queries related to why loan is required by a person and is answered in 4 minutes.
- **Activity 2**: The SBI counter person gathers all the related information from the customer on an average in 5 minutes.

• **Activity 3**: The SBI counter person analyzes the information and checks the papers/documents, etc. of the customer to decide the loan and amount and on an average it takes 7 minutes. • **Activity 4**: The SBI counter person finally checks and verifies all the papers/documents and on an average it takes 2 minutes. • **Activity 5**: Finally the customer signs the documents and has the loan money transferred to his/her account and on an average it takes 4 minutes. The whole process is conducted by three (03) SBI persons in a worker-paced line. The assignment of tasks to SBI personnel is as follows: **W1 does activity 1**, **W2 does activities 2 and 3**, and **W3 does activities 4 and 5**. Assume each SBI person is paid the same wages which is Rs. 2000/- per hour. You can assume that there exists **unlimited demand** and that loans are only entering the process at the rate of the bottleneck. Then what is cost of direct labour (in Rs. per loan)? **ANSWER IS IN INTEGER**

Range	1119 to 1201
Given unlimited demand, we get a flow rate which is determined by the capacity of the process, which is 1/12 per minute = 60×(1/12) = 5 loan per hour. Costs of direct labor =Wages/Flow Rate = (2000×3)/5 = Rs. 1200/- per loan	

A primary hospital has the **capacity** to see 75 patients per 1 hour. The average **demand** rate is 385 patients per 5 hours. What is the utilization in percentage for the hospital? **ANSWER IS IN INTEGER**

Range	99.9 to 100.1
The calculations are shown below - Capacity is 5×75 = 375 patients per 5 hours - Demand is 385 patients per 5 hours - Flow Rate = Minimum {Demand, Process Capacity} = Min(375, 385) = 375 patients per 5 hours - Utilization = {Flow Rate/Capacity} = (375/375) = 1.00 which is 100%	

Given P = Rs. 100/-, r = 10% pa (Simple Interest Rate) and n = 1 yr, the amount (in Rs.) after 1 year is? **ANSWER IS IN INTEGER**

Range	99.9 to 100.1

$A = P(1 + r/100)^n = 100(1 + 0.1)^1 = 110$ (in Rs.)

For the LPP formulation $Max\ f(\boldsymbol{x}) \equiv Max\ f(x_1, x_2) \equiv Max{:}\ 3x_1 + 9x_2$ s.t.: $x_1 + 4x_2 \leq 8$ $x_1 + 2x_2 \leq 4$ $x_1, x_2 \geq 0$ The solution(s) is/are	
A	Degenerate
B	Unique and optimal
C	Unbounded
D	Alternative or Multiple Optimal
E	Infeasible
Degenerate	

For the LPP formulation $Max\ f(\boldsymbol{x}) \equiv Max\ f(x_1, x_2) \equiv Max{:}\ 3x_1 - x_2$ s.t.: $2x_1 + x_2 \geq 2$ $x_1 + 3x_2 \leq 3$ $x_2 \leq 4$ $x_1, x_2 \geq 0$ The solution(s) is/are	
A	Degenerate
B	Unique and optimal
C	Unbounded
D	Alternative or Multiple Optimal
E	Infeasible

Unique and optimal $(x^*_1, x^*_2) = (+3, 0)$ and $f(\boldsymbol{x}^*) = f(x^*_1, x^*_2) = f(+3, 0) = 3\times3 - 1\times0 = 9$

For the LPP formulation $Max\ f(\boldsymbol{x}) \equiv Max\ f(x_1, x_2) \equiv Max: +5x_1 + x_2$ s.t.: $2x_1 + 3x_2 \geq 6$ $3x_1 + 2x_2 \geq 6$ $x_1, x_2 \geq 0$ The solution(s) is/are	
A	Degenerate
B	Unique and optimal
C	Unbounded
D	Alternative or Multiple Optimal
E	Infeasible
Unbounded $(x^*_1, x^*_2) = (+\infty, +\infty)$ and $f(\boldsymbol{x}^*) = f(x^*_1, x^*_2) = = f(+\infty, +\infty) = +5\times\infty + \infty = +\infty$	

For the LPP formulation $Min\ x\ f(\boldsymbol{x}) \equiv Min\ f(x_1, x_2) \equiv Min: +4x_1 + 2x_2$ s.t.: $2x_1 + x_2 \geq 10$ $x_1 + 2x_2 \geq 8$ $x_1, x_2 \geq 0$ The solutions is/are	
A	Degenerate
B	Unique and optimal
C	Unbounded
D	Alternative or Multiple Optimal

E	Infeasible
Alternative or Multiple Optimal $(x^*_1, x^*_2) = (0, 10)$ and $f(\boldsymbol{x}^*) = f(x^*_1, x^*_2) = f(0, +10) = 4\times0 + 2\times10 = +20$ $(x^*_1, x^*_2) = (+4, +2)$ and $f(\boldsymbol{x}^*) = f(x^*_1, x^*_2) = f(+4, +2) = 4\times4 + 2\times2 = +20$	

For the LPP formulation $Max\ f(\boldsymbol{x}) \equiv Max\ f(x_1, x_2) \equiv Max: +5x_1 +4x_2$ s.t.: $2x_1 + 3x_2 \leq 6$ $5x_1 + 4x_2 \geq 20$ $x_1, x_2 \geq 0$ The solution(s) is/are	
A	Degenerate
B	Unique and optimal
C	Unbounded
D	Alternative or Multiple Optimal
E	Infeasible
Infeasible (x^*_1, x^*_2) = Does not exist as there is no feasible region	